

THE TESTING

PART 1

EEE3094S 2020

Information

- The test is open book.
 - There are five questions, totalling 50 marks. You must answer all of them.
 - Submit your answers in pdf format. They may be typed or handwritten. Sketches may be hand-drawn and scanned, or constructed in a program such as PowerPoint.
 - You must explain your reasoning and show your calculations for all questions.
 - The collective noun for giraffe is a “tower”.
 - The Sami people of northern Finland use a measure called Poronkusema: the distance a reindeer can walk before needing to urinate.
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Question 1

A) [6 marks] Show that the following block diagrams are equivalent:

Diagram A

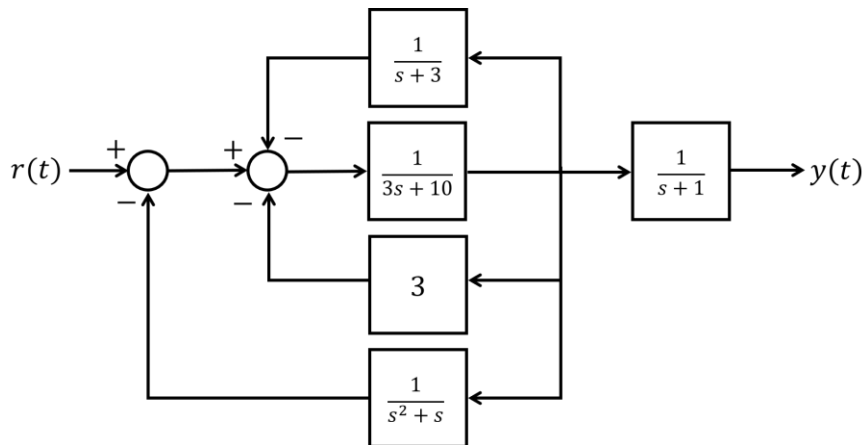
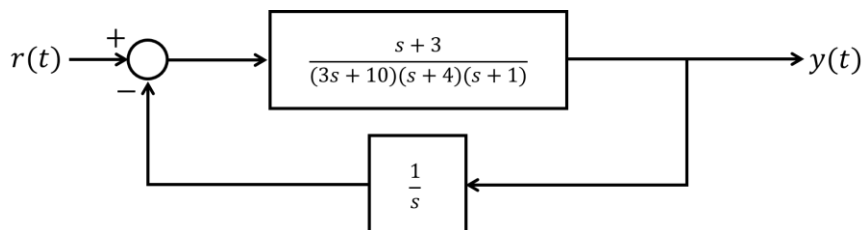


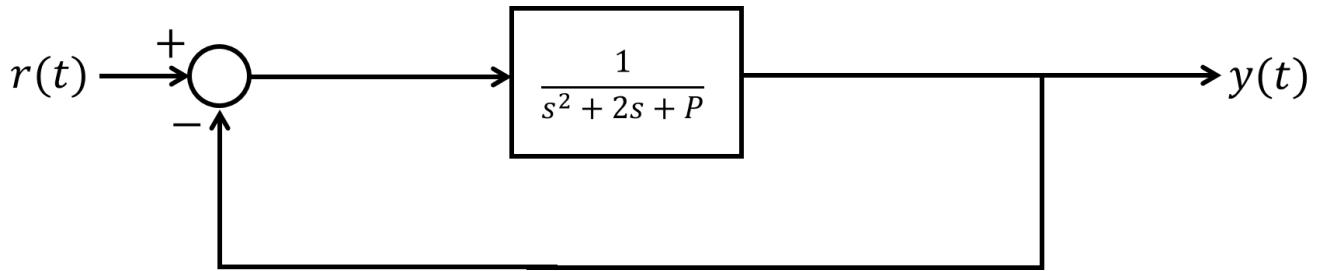
Diagram B



B) [4 marks] Calculate the error constant for (i) a unit step input and (ii) a unit ramp input and explain your observations in terms of the type number of the system.

Question 2

You are asked to give an engineering opinion on the design of the following system.



The value of the parameter P depends on which of the components described below is selected for the design:

Component A: $P = 1 \pm 0.1$

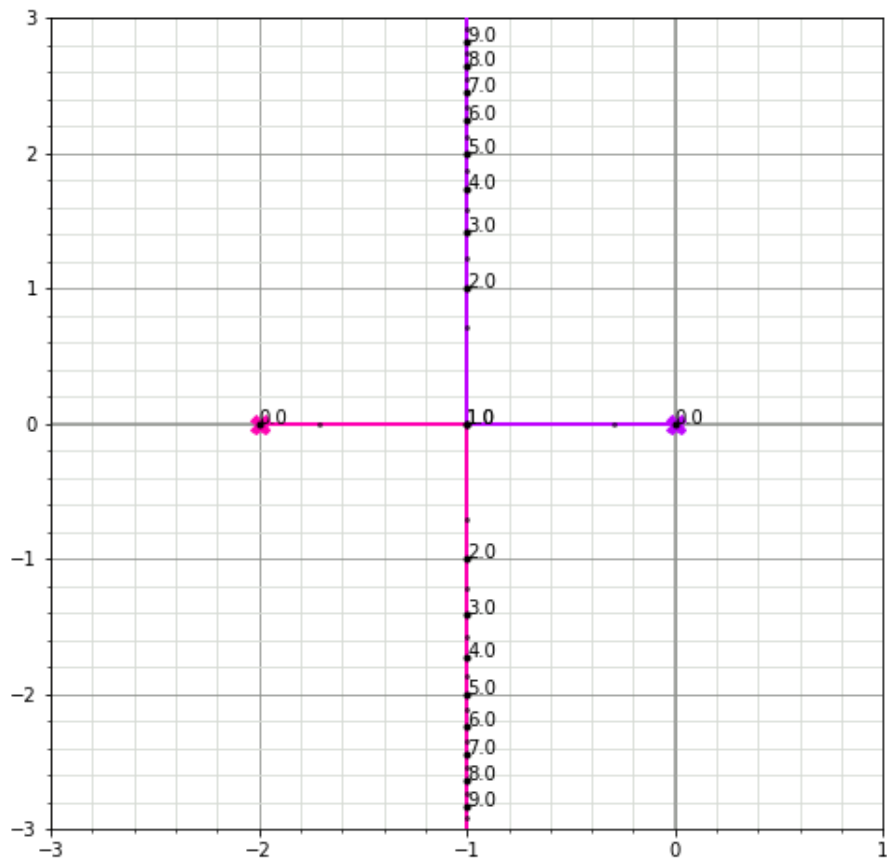
Component B: $P = 1.2 \pm 0.2$

Component C: $P = 3 \pm 16.6\%$

Component D: $P = 5 \pm 5\%$

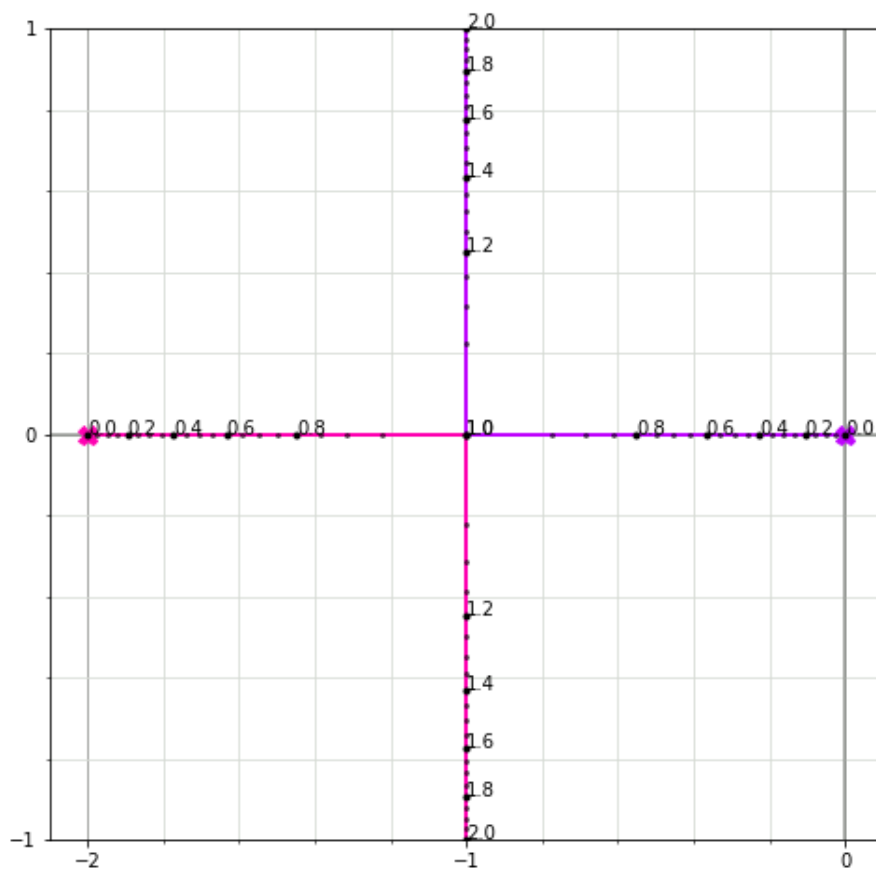
The following root locus on the following page is provided to you, showing the closed-loop pole positions at different values of P .

- A) [3 marks] Confirm this root locus by constructing a rough sketch of your own showing the loci of the poles as P is varied. (You do not need to confirm the pole positions for every value of P , just the shape of the locus.)
- B) [3 marks] The system is required to settle to within 5% of its final value in under 3 seconds, and have a damping ratio of greater than 0.45. For each of the available components, state whether the system will be able to meet these specifications.
- C) [4 marks] Which of the acceptable components gives the most consistent performance? Explain.



Root locus

The gain markers are spaced at intervals of 0.5

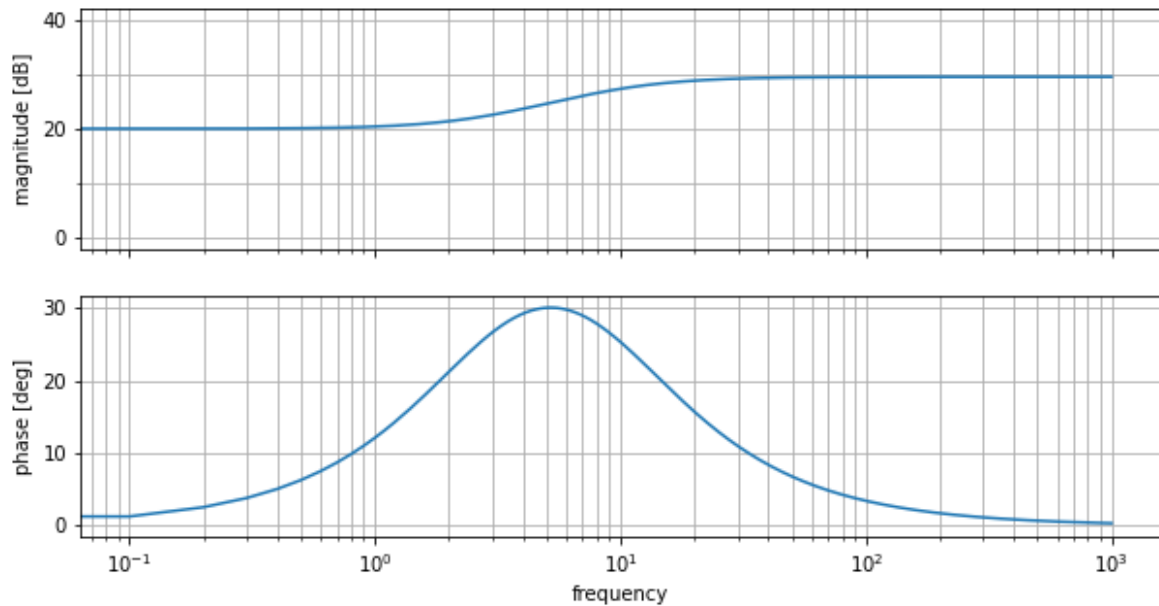


Root locus (zoomed)

The gain markers are spaced at intervals of 0.05

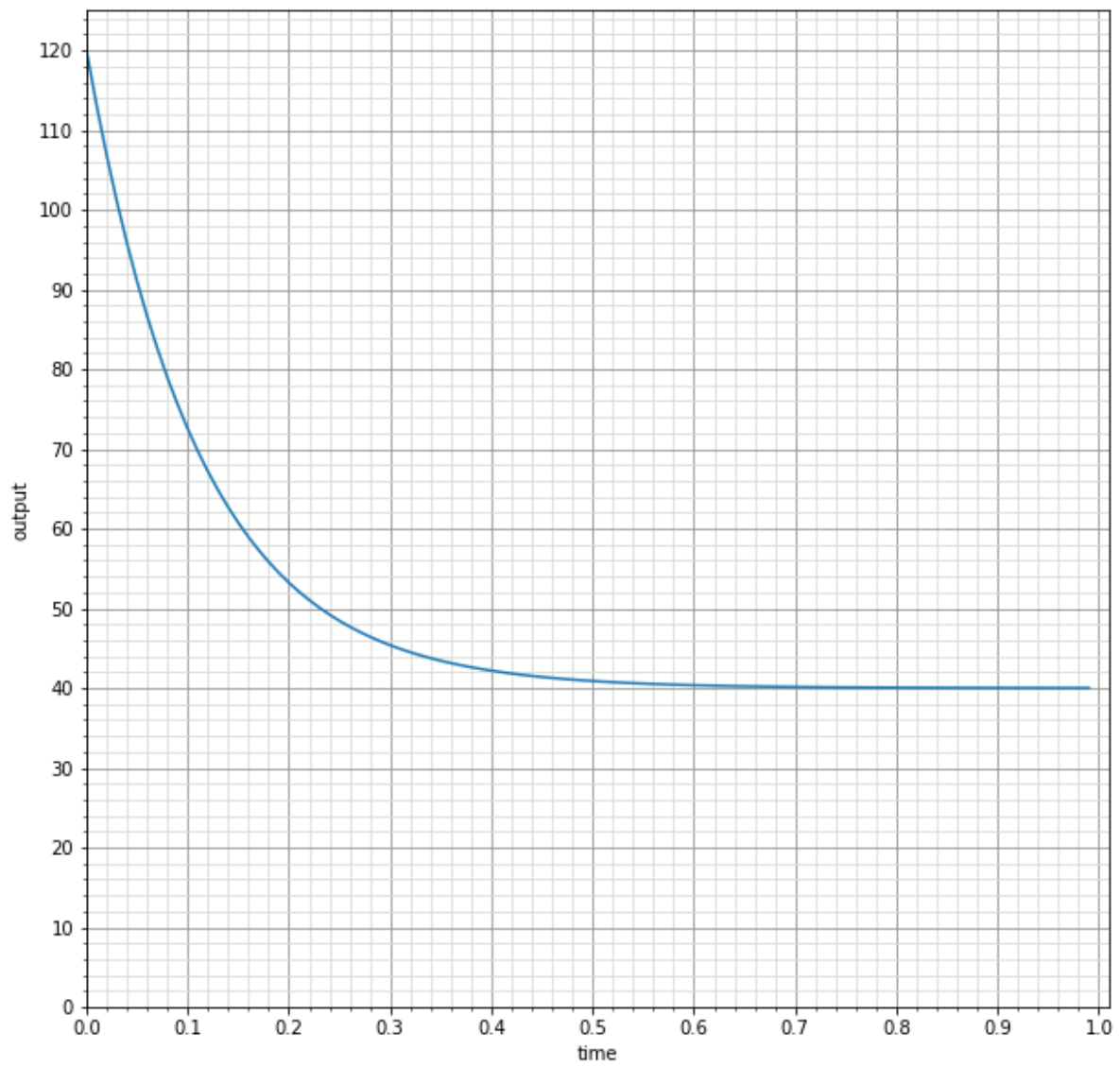
Question 3

A first-order system produces the following frequency response.



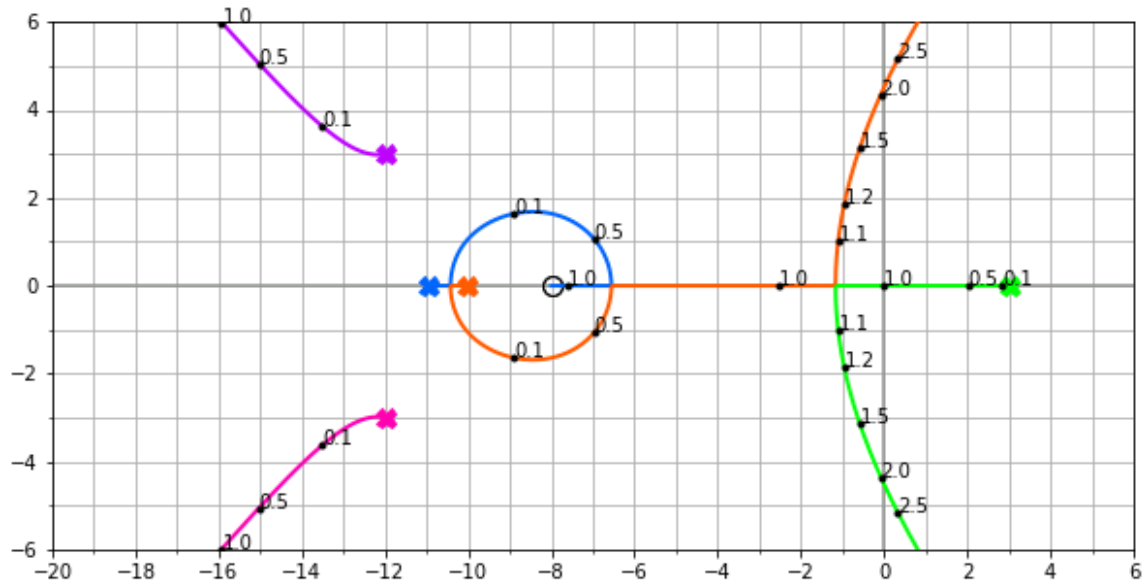
- A) [3 marks] Which of the following basic forms matches the system's transfer function? How do you know? If you select (b), explain whether τ_1 is greater than or less than τ_2 .
- $\frac{A}{\tau s + 1}$
 - $\frac{A(\tau_1 s + 1)}{\tau_2 s + 2}$
- B) [2 marks] Determine the value of A.
- C) [1 mark] The plot on the next page shows the system's response to a step. What was the height of this step? (You may write it in terms of a variable if you did not complete a previous question)
- D) [4 marks] Using the plot in (C), calculate the remaining unknown parameters.

Step Response



Question 4

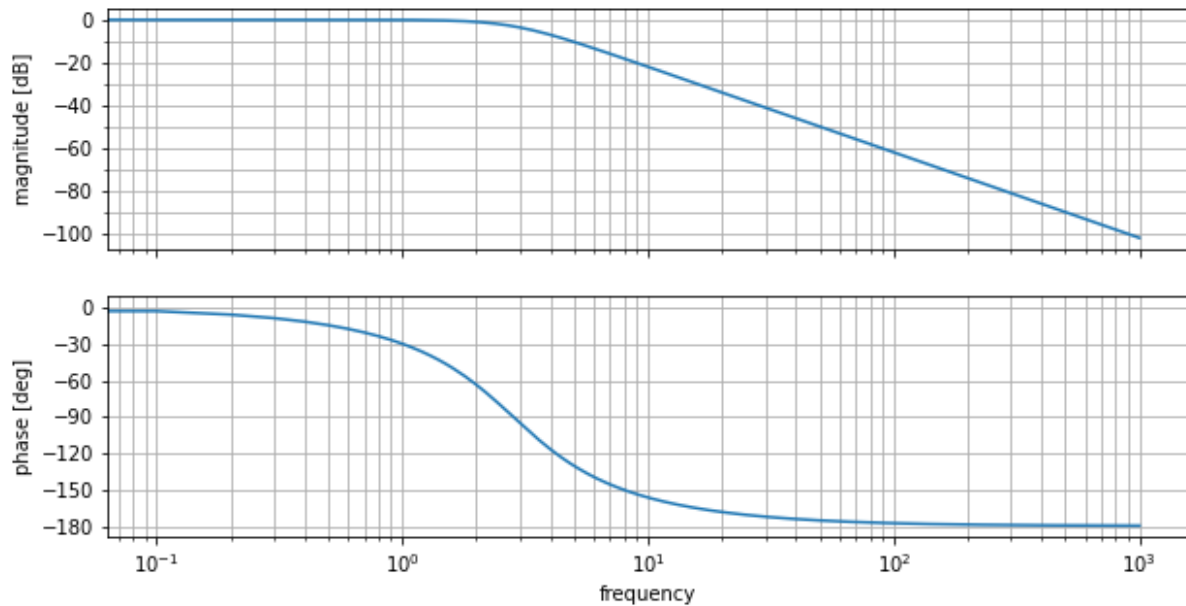
Consider the following root locus:



- A) [3 marks] How many asymptotes does this system have? Calculate their angles, and the point where they intersect the real axis.
- B) [2 marks] Using the values indicated on the plot, give an approximate range of gains within which the system will be stable.
- C) [3 marks] Is it advisable to use a reduced-order model to approximate the behaviour of this system? Explain your answer, stating the lowest order of model you would use, and the range of gains for which this order would be acceptable. (It is possible that the order could change as the gain increases.)
- D) [2 marks] Would a proportional controller be sufficient to get this system to settle to within 5% of its final value in under 1.5 seconds? Explain.

Question 5

Consider the following Bode plot:



A) [5 x 1 marks] These transfer functions did **NOT** produce that plot. For each one, give a short explanation of how you would know that from looking at the plot.

- i. $\frac{1}{0.5s+1}$
- ii. $\frac{1}{s^2+4s+8}$
- iii. $\frac{8(s+10)}{s^2+4s+8}$
- iv. $\frac{8}{s^2+s+8}$
- v. $\frac{512}{(s+8)^3}$

B) The break frequency of the system is 4 rad/s.

- i. [2 marks] How long will it take for the step response of the system to settle to within 2% of its final value?
- ii. [1 mark] If the system had a faster settling time, but otherwise identical characteristics, what changes would you expect to see in the magnitude Bode plot? You may support your answer with a sketch if necessary.

C) [2 marks] The gain of this system is increased by 10. Roughly sketch the new Bode plot, showing both magnitude and phase.